

Prakhar Shekhar Parthasarthi

Software Engineer · Distributed systems · Java 17 · AWS

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Shahjahanpur, UP · open to relocate

github.com/praxstack | linkedin.com/in/prakharshekhar | prax-portfolio-one.vercel.app

47%

p90 latency cut

195k

GET rps · Java Redis

4

services owned

80%+

line coverage

SUMMARY

Software Development Engineer, **3+ years at Amazon** Travel. Owned **4 microservices** in **Java 17 / Spring Boot**: `CompletableFuture` fan-out, idempotent **SQS** (43 bps CPT), and **CDK (Java)**. p90 cut 47% (1.1s → 0.59s), conversion +33%, coverage 80%+.

SKILLS

Java & JVM	Java 17, Spring Boot/MVC, Guice, JUnit 5, Mockito, Maven · concurrency, GC profiling
Distributed	Microservices, REST, GraphQL, idempotency, state machines, DLQ, at-least-once, TDD
AWS	Lambda, DynamoDB, SQS, Step Functions, API Gateway, CloudWatch, ECS, CDK (Java)
Data	DynamoDB, Redis, NoSQL modelling, SQL, caching
Front-end	React (SSR & CSR), Redux, HTML/CSS, TypeScript
Tooling	Git, CI/CD, JaCoCo gates, Agile, on-call, incident response · Python, C++

EXPERIENCE

Software Development Engineer, Jul 2022 — Sep 2025

Amazon · Travel (Hotels & Flights) · Bengaluru

- Ownership. Owned **4 customer-facing microservices** (search → order review); Spring Boot REST, DynamoDB DAOs, Prime and partner discount flows — ~30 bps funnel churn drop.
- Concurrency. Parallelised 6 partner APIs via `CompletableFuture.allOf()` — p90 47% (1.1s→0.59s), conversion +33%; GZIP cut wire size 95%.
- Event-driven. Designed an idempotent flight schedule-change webhook (SQS, dedup keys) — 43 bps CPT; Step Functions booking state machine with Java Lambdas.
- IaC. Authored service infra in **Java CDK**; migrated 2 services from CloudFormation — rollback incidents halved.
- Testing. Held 80%+ line coverage; `@SpringBootTest` + in-memory DynamoDB; introduced Gradle JaCoCo gates in CI.

Software Development Engineer, Intern, May 2021 – Jul 2021 · Feb 2022 – Jul 2022

Amazon · Bengaluru

- Migrated GraphQL resolvers to Java + DynamoDB (API Gateway + Lambda); shipped **One-Click Renewal**; 90%+ unit-test coverage (JUnit 5 + Mockito).

PROJECTS

Redis Server in Java

github.com/praxstack/redis-server-java

RESP2 from scratch; bounded thread pool, `ConcurrentHashMap`, hybrid TTL. GET 195k rps, SET 154k rps, p50 < 0.3 ms; 42 tests.

Open Source — Fabric (42k stars)

danielmiessler/Fabric · Go

3 merged PRs: Bedrock bearer auth (#2044), dynamic regions (#2052), streaming deadlock fix (#2061).

AI Visual Code Review

github.com/praxstack/ai-visual-code-review · npm v2.4.1

Visual diff review + VS Code extension; TypeScript, Jest, command-injection hardening, AI markdown exports.

Audio Transcription Pipeline

github.com/praxstack/audio-transcription-pipeline

Six-stage audio/video → speaker-labelled Markdown. Bedrock Voxtral ASR + Claude repair. 63 tests.

EDUCATION

Master of Computer Applications (M.C.A.), Computer Science · MNNIT, Prayagraj · GPA 9.25 · 2019 — 2022

Bachelor of Science (B.Sc. Hons), Computer Science · University of Delhi · 2015 — 2018

AWARDS

Amazon Pay Merchants Categories Champion — 100+ hotel trust-buster bugs; on-call 20+ to ~5 per shift · 2023